

DM545/DM871
Linear and Integer Programming

Duality

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1. Derivation and Motivation

2. Theory

1. Derivation and Motivation

2. Theory

Dual variables $\mathbf{y}$ in one-to-one correspondence with the constraints:

Primal problem:

$$\begin{aligned}\max \quad & z = \mathbf{c}^T \mathbf{x} \\ & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{aligned}$$

Dual Problem:

$$\begin{aligned}\min \quad & w = \mathbf{b}^T \mathbf{y} \\ & A^T \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq \mathbf{0}\end{aligned}$$

Bounding approach

$$\begin{aligned} z^* = \max \quad & 4x_1 + x_2 + 3x_3 \\ & x_1 + 4x_2 \leq 1 \\ & 3x_1 + x_2 + x_3 \leq 3 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

a feasible solution is a **lower bound** but how good?

By tentatives:

$$(x_1, x_2, x_3) = (1, 0, 0) \rightsquigarrow z^* \geq 4$$

$$(x_1, x_2, x_3) = (0, 0, 3) \rightsquigarrow z^* \geq 9$$

What about **upper bounds**?

$$\begin{array}{rcl} & 2 \cdot (x_1 + 4x_2) & \leq 2 \cdot 1 \\ & + 3 \cdot (3x_1 + x_2 + x_3) & \leq 3 \cdot 3 \\ \hline 4x_1 + x_2 + 3x_3 & \leq & 11x_1 + 11x_2 + 3x_3 \leq 11 \end{array}$$
$$\mathbf{c}^T \mathbf{x} \leq \mathbf{y}^T A \mathbf{x} \leq \mathbf{y}^T \mathbf{b}$$

Hence $z^* \leq 11$. Is this the best upper bound we can find?

multipliers $y_1, y_2 \geq 0$ that preserve sign of inequality

$$\begin{array}{rcl} y_1 \cdot (x_1 + 4x_2) & \leq & y_1(1) \\ y_2 \cdot (3x_1 + x_2 + x_3) & \leq & y_2(3) \\ \hline (y_1 + 3y_2)x_1 + (4y_1 + y_2)x_2 + y_2x_3 & \leq & y_1 + 3y_2 \end{array}$$

Coefficients

$$\begin{array}{rcl} y_1 + 3y_2 & \geq & 4 \\ 4y_1 + y_2 & \geq & 1 \\ y_2 & \geq & 3 \end{array}$$

$z = 4x_1 + x_2 + 3x_3 \leq (y_1 + 3y_2)x_1 + (4y_1 + y_2)x_2 + y_2x_3 \leq y_1 + 3y_2$
then to attain the best upper bound:

$$\begin{array}{rcl} \min & y_1 + 3y_2 \\ & y_1 + 3y_2 \geq 4 \\ & 4y_1 + y_2 \geq 1 \\ & y_2 \geq 3 \\ & y_1, y_2 \geq 0 \end{array}$$

$$\begin{array}{rcl}
 -z = & \pi_1 b_1 & + \pi_2 b_2 \dots + \pi_m b_m \\
 & \pi_1 a_{11} & + \pi_2 a_{21} \dots + \pi_m a_{m1} \leq -c_1 \\
 & \vdots & \ddots \qquad \qquad \qquad \vdots \\
 & \pi_1 a_{1n} & + \pi_2 a_{2n} \dots + \pi_m a_{mn} \leq -c_n \\
 & & \pi_1, \pi_2, \dots, \pi_m \leq 0
 \end{array}$$

$$\mathbf{y} = -\boldsymbol{\pi}$$

$$\begin{array}{rcl}
 -z = & (-y_1 b_1) & + (-y_2 b_2) \dots + (-y_m b_m) \\
 & (-y_1 a_{11}) & + (-y_2 a_{21}) \dots + (-y_m a_{m1}) \leq -c_1 \\
 & \vdots & \ddots \qquad \qquad \qquad \vdots \\
 & (-y_1 a_{1n}) & + (-y_2 a_{2n}) \dots + (-y_m a_{mn}) \leq -c_n \\
 & & -y_1, -y_2, \dots, -y_m \leq 0
 \end{array}$$

as we will see $\mathbf{b}^T \mathbf{y} \geq \mathbf{c}^T \mathbf{x}$, hence it is more interesting to minimize. It yields:

$$\begin{array}{ll}
 \min & \mathbf{b}^T \mathbf{y} \\
 & A^T \mathbf{y} \geq \mathbf{c} \\
 & \mathbf{y} \geq \mathbf{0}
 \end{array}$$

Example

$$\begin{array}{rcll} \max & 6x_1 & + & 8x_2 \\ & 5x_1 & + & 10x_2 \leq 60 \\ & 4x_1 & + & 4x_2 \leq 40 \\ & & & x_1, x_2 \geq 0 \end{array}$$

$$\left\{ \begin{array}{rcll} 5\pi_1 & + & 4\pi_2 & + 6\pi_3 \leq 0 \\ 10\pi_1 & + & 4\pi_2 & + 8\pi_3 \leq 0 \\ 1\pi_1 & + & 0\pi_2 & + 0\pi_3 \leq 0 \\ 0\pi_1 & + & 1\pi_2 & + 0\pi_3 \leq 0 \\ 0\pi_1 & + & 0\pi_2 & + 1\pi_3 = 1 \\ 60\pi_1 & + & 40\pi_2 & \end{array} \right.$$

$$y_1 = -\pi_1 \geq 0$$

$$y_2 = -\pi_2 \geq 0$$

Duality Recipe

	Primal linear program	Dual linear program
Variables	$x_1, x_2, \dots, x_n$	$y_1, y_2, \dots, y_m$
Matrix	A	A^T
Right-hand side	$\mathbf{b}$	$\mathbf{c}$
Objective function	$\max \mathbf{c}^T \mathbf{x}$	$\min \mathbf{b}^T \mathbf{y}$
Constraints	i th constraint has $\leq$ $\geq$ $=$	$y_i \geq 0$ $y_i \leq 0$ $y_i \in \mathbb{R}$
	$x_j \geq 0$ $x_j \leq 0$ $x_j \in \mathbb{R}$	j th constraint has $\geq$ $\leq$ $=$

1. Derivation and Motivation

2. Theory

The dual of the dual is the primal:

Primal Problem:

$$\begin{aligned}\max \quad & z = c^T x \\ & Ax \leq b \\ & x \geq 0\end{aligned}$$

Let's put the dual in the standard form

Dual Problem:

$$\begin{aligned}\min \quad & b^T y \equiv -\max -b^T y \\ & -A^T y \leq -c \\ & y \geq 0\end{aligned}$$

Dual Problem:

$$\begin{aligned}\min \quad & w = b^T y \\ & A^T y \geq c \\ & y \geq 0\end{aligned}$$

Dual of Dual:

$$\begin{aligned}-\min \quad & -c^T x \\ & -Ax \geq -b \\ & x \geq 0\end{aligned}$$

Weak Duality Theorem

As we saw the dual produces upper bounds. This is true in general:

Theorem (Weak Duality Theorem)

Given:

$$(P) \max\{\mathbf{c}^T \mathbf{x} \mid A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$$

$$(D) \min\{\mathbf{b}^T \mathbf{y} \mid A^T \mathbf{y} \geq \mathbf{c}, \mathbf{y} \geq \mathbf{0}\}$$

for any feasible solution $\mathbf{x}$ of (P) and any feasible solution $\mathbf{y}$ of (D):

$$\mathbf{c}^T \mathbf{x} \leq \mathbf{b}^T \mathbf{y}$$

Proof:

From (D) $c_j \leq \sum_{i=1}^m y_i a_{ij} \forall j$ and from (P) $\sum_{j=1}^n a_{ij} x_j \leq b_i \forall i$

From (D) $y_i \geq 0$ and from (P) $x_j \geq 0$

$$\sum_{j=1}^n c_j x_j \leq \sum_{j=1}^n \left(\sum_{i=1}^m y_i a_{ij} \right) x_j = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j \right) y_i \leq \sum_{i=1}^m b_i y_i$$

Strong Duality Theorem

Due to Von Neumann and Dantzig 1947 and Gale, Kuhn and Tucker 1951.

Theorem (Strong Duality Theorem)

Given:

$$(P) \max\{\mathbf{c}^T \mathbf{x} \mid A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$$

$$(D) \min\{\mathbf{b}^T \mathbf{y} \mid A^T \mathbf{y} \geq \mathbf{c}, \mathbf{y} \geq \mathbf{0}\}$$

exactly one of the following occurs:

1. *(P) and (D) are both infeasible*
2. *(P) is unbounded and (D) is infeasible*
3. *(P) is infeasible and (D) is unbounded*
4. *(P) has feasible solution, then let an optimal be: $\mathbf{x}^* = [x_1^*, \dots, x_n^*]$
(D) has feasible solution, then let an optimal be: $\mathbf{y}^* = [y_1^*, \dots, y_m^*]$, then:*

$$\mathbf{c}^T \mathbf{x}^* = \mathbf{b}^T \mathbf{y}^*$$

Proof:

- all other combinations of 3 possibilities (Optimal, Infeasible, Unbounded) for (P) and 3 for (D) are ruled out by weak duality theorem.
- we use the simplex method. (Other proofs independent of the simplex method exist, eg, Farkas Lemma and convex polyhedral analysis)
- The last row of the final tableau will give us that for every $\mathbf{x}$:

$$\begin{aligned} z &= z^* + \sum_{k=1}^{n+m} \bar{c}_k x_k = z^* + \sum_{j=1}^n \bar{c}_j x_j + \sum_{i=1}^m \bar{c}_{n+i} x_{n+i} \\ &= z^* + \bar{\mathbf{c}}_B \mathbf{x}_B + \bar{\mathbf{c}}_N \mathbf{x}_N \end{aligned} \quad (*)$$

In addition, $z^* = \sum_{j=1}^n c_j x_j^*$ (c_j , original values) because optimal value

- We define $y_i^* = -\bar{c}_{n+i}, \quad i = 1, 2, \dots, m$

- We claim that $(y_1^*, y_2^*, \dots, y_m^*)$ is a dual feasible solution satisfying $\mathbf{c}^T \mathbf{x}^* = \mathbf{b}^T \mathbf{y}^*$.

- Let's verify the claim:

We substitute in (*): i) $z = \sum_{j=1}^n c_j x_j$; ii) $\bar{c}_{n+i} = -y_i^*$; and iii) $x_{n+i} = b_i - \sum_{j=1}^n a_{ij} x_j$ for $i = 1, 2, \dots, m$ ($n+i$ are the slack variables)

$$\begin{aligned} \sum_{j=1}^n c_j x_j &= z^* + \sum_{j=1}^n \bar{c}_j x_j - \sum_{i=1}^m y_i^* \left(b_i - \sum_{j=1}^n a_{ij} x_j \right) \\ &= \left(z^* - \sum_{i=1}^m y_i^* b_i \right) + \sum_{j=1}^n \left(\bar{c}_j + \sum_{i=1}^m a_{ij} y_i^* \right) x_j \end{aligned}$$

This must hold for every $(x_1, x_2, \dots, x_n)$ hence:

$$\begin{aligned} z^* &= \sum_{i=1}^m b_i y_i^* & \implies y^* \text{ satisfies } c^T x^* = b^T y^* \\ c_j &= \bar{c}_j + \sum_{i=1}^m a_{ij} y_i^*, j = 1, 2, \dots, n \end{aligned}$$

Since $\bar{c}_k \leq 0$ for every $k = 1, 2, \dots, n + m$:

$$\bar{c}_j \leq 0 \rightsquigarrow c_j - \sum_{i=1}^m y_i^* a_{ij} \leq 0 \rightsquigarrow$$

$$\sum_{i=1}^m y_i^* a_{ij} \geq c_j \quad j = 1, 2, \dots, n$$

$$\bar{c}_{n+i} \leq 0 \rightsquigarrow y_i^* = -\bar{c}_{n+i} \geq 0,$$

$$i = 1, 2, \dots, m$$

$\Rightarrow y^*$ is also dual feasible solution

Complementary Slackness Theorem

Theorem (Complementary Slackness)

A feasible solution $\mathbf{x}^*$ for (P)

A feasible solution $\mathbf{y}^*$ for (D)

Necessary and sufficient conditions for optimality of both:

$$\left(c_j - \sum_{i=1}^m y_i^* a_{ij} \right) x_j^* = 0, \quad j = 1, \dots, n$$

If $x_j^* \neq 0$ then $\sum y_i^* a_{ij} = c_j$ (no surplus)

If $\sum y_i^* a_{ij} > c_j$ then $x_j^* = 0$

Proof:

$$\mathbf{z}^* = \mathbf{c}^T \mathbf{x}^* \leq \mathbf{y}^* \mathbf{A} \mathbf{x}^* \leq \mathbf{b}^T \mathbf{y}^* = \mathbf{w}^*$$

Hence from strong duality theorem:

$$\mathbf{c} \mathbf{x}^* - \mathbf{y}^* \mathbf{A} \mathbf{x}^* = 0$$

In scalars

$$\sum_{j=1}^n \underbrace{\left(c_j - \sum_{i=1}^m y_i^* a_{ij} \right)}_{\leq 0} \underbrace{x_j^*}_{\geq 0} = 0$$

Hence each term must be $= 0$

Proof in scalar form:

$$c_j x_j^* \leq \left(\sum_{i=1}^m a_{ij} y_i^* \right) x_j^* \quad j = 1, 2, \dots, n \quad \text{from feasibility in D}$$

$$\left(\sum_{j=1}^n a_{ij} x_j^* \right) y_i^* \leq b_i y_i^* \quad i = 1, 2, \dots, m \quad \text{from feasibility in P}$$

Summing in j and in i :

$$\sum_{j=1}^n c_j x_j^* \leq \sum_{j=1}^n \left(\sum_{i=1}^m a_{ij} y_i^* \right) x_j^* = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j^* \right) y_i^* \leq \sum_{i=1}^m b_i y_i^*$$

For the strong duality theorem the left hand side is equal to the right hand side and hence all inequalities become equalities.

$$\sum_{j=1}^n \underbrace{\left(c_j - \sum_{i=1}^m y_i^* a_{ij} \right)}_{\leq 0} \underbrace{x_j^*}_{\geq 0} = 0$$

Economic Interpretation of Duality Theory

$$\begin{aligned}
 \max \quad & 5x_0 + 6x_1 + 8x_2 \\
 & 6x_0 + 5x_1 + 10x_2 \leq 60 \\
 & 8x_0 + 4x_1 + 4x_2 \leq 40 \\
 & 4x_0 + 5x_1 + 6x_2 \leq 50 \\
 & x_0, x_1, x_2 \geq 0
 \end{aligned}$$

final tableau:

$$\begin{array}{ccccccc|c}
 x_0 & x_1 & x_2 & s_1 & s_2 & s_3 & -z & b \\
 \hline
 & 0 & 1 & & 0 & & & 5/2 \\
 & 1 & 0 & & 0 & & & 7 \\
 & 0 & 0 & & 1 & & & 2 \\
 \hline
 -1/5 & 0 & 0 & -1/5 & 0 & -1 & & -62
 \end{array}$$

- What is the maximum number of products that can be produced? at most m
- What are the values of the variables, the reduced costs, the dual variables, and the shadow (or marginal) prices?
- What is the value of additional manpower capacity (1st constr.)? In +1 out +1/5
- If one slack variable > 0 then overcapacity: $s_2 = 2$ then the second constraint is not tight
- How much more expensive would a non-selected product need to be? look at reduced costs:

$$c_j + \pi a_j > 0$$

Economic Interpretation of Duality Theory

Game: Suppose two economic operators:

- P owns the factory and produces goods
- D is in the market buying and selling raw material and resources
- D asks P to close and sell him/her all resources
- P considers if the offer is convenient
- D wants to spend the least possible
- y are prices that D offers for the resources
- $\sum y_i b_i$ is the amount D has to pay to have all resources of P
- $\sum y_i a_{ij} \geq c_j$ the total value to make j is greater or eq. than the price per unit of product j
- P either sells all resources $\sum y_i a_{ij}$ or produces product j (c_j)
- without $\geq$ there would not be negotiation because P would be better off producing and selling
- ▶ at optimality the situation is indifferent (strong th.)
- ▶ resource 2 that was not totally utilized in the primal has been given value 0 in the dual.
(complementary slackness th.) Plausible, since we do not use all the resource, likely to place not so much value on it.
- ▶ for product 0 $\sum y_i a_{ij} > c_j$ hence not profitable producing it. (complementary slackness th.)

- Derivation:
 - Economic Interpretation
 - Bounding Approach
 - Multiplier Approach
 - Recipe
 - Lagrangian Multipliers Approach (next time)
- Theory:
 - Symmetry
 - Weak Duality Theorem
 - Strong Duality Theorem
 - Complementary Slackness Theorem